

LAB 14

(Total Marks 10)

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01/12/2023

Task 1: Write a program that detects if a character entered by the user is his/her full name or not using hardware interrupt. And Display Y in case of yes else N.

[org 0x100]

jmp start

oldisr: dd 0 ; space for saving old isr

 ; keyboard interrupt service routine

kbisr:

 push ax

 push es

 mov ax, 0xb800

 mov es, ax ; point es to video memory

 in al, 0x60 ; read a char from keyboard port

 cmp al, 0x32 ;M

 je print

 cmp al, 0x16 ;U

 je print

 cmp al, 0x23 ;H

 je print

 cmp al, 0x1E ;A

 je print

 cmp al, 0x20 ;D

 je print

 cmp al, 0x14 ;T

 je print

 mov byte [es:0], 'N'

 jmp nomatch

print:

 mov byte [es:0], 'Y'

nomatch:

 ;EOI is no longer needed as the original BIOS routine will have it at its end.

 ;mov al, 0x20

 ;out 0x20, al

 pop es

 pop ax

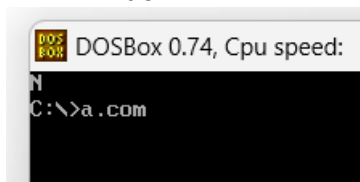
```
jmp far [cs:oldisr] ; call the original ISR  
; iret
```

start:

```
xor ax, ax  
mov es, ax ; point es to IVT base  
mov ax, [es:9*4]  
mov [oldisr], ax ; save offset of old routine  
mov ax, [es:9*4+2]  
mov [oldisr+2], ax ; save segment of old routine  
cli ; disable interrupts  
mov word [es:9*4], kbisr ; store offset at n*4  
mov [es:9*4+2], cs ; store segment at n*4+2  
sti ; enable interrupts
```

l1:

```
mov ah, 0 ; service 0 – get keystroke  
int 0x16 ; call BIOS keyboard service  
cmp al, 27 ; is the Esc key pressed  
jne l1 ; if no, check for next key  
mov ax, [oldisr] ; read old offset in ax  
mov bx, [oldisr+2] ; read old segment in bx  
cli ; disable interrupts  
mov [es:9*4], ax ; restore old offset from ax  
mov [es:9*4+2], bx ; restore old segment from bx  
sti ; enable interrupts  
mov ax, 0x4c00 ; terminate program  
int 0x21
```



Task 2: Write a program that displays an indicator on the screen when the CAPS Lock key is pressed using hardware interrupt. The indicator should toggle between 'ON' and 'OFF' with each press.

```
[org 0x100]
jmp start
oldisr: dd 0 ; space for saving old isr
        ; keyboard interrupt service routine
kbisr:
    push ax
    push es

    mov ax, 0xb800
    mov es, ax ; point es to video memory

    in al, 0x60 ; read a char from keyboard port

    cmp al, 0x3a    ;caps lock is down
    je pressed

    cmp al, 0xba    ;caps is up
    je release

    jne nomatch

release:
    mov byte [es:0], 'O'
    mov byte [es:2], 'F'
    mov byte [es:4], 'F'
    jmp nomatch

pressed:
    mov byte [es:0], 'O'
    mov byte [es:2], 'N'
    mov byte [es:4], ' ' ;removing (F) of (OFF)
nomatch:
    ;EOI is no longer needed as the original BIOS routine will have it at its end.
    ;mov al, 0x20
    ;out 0x20, al

    pop es
    pop ax
    jmp far [cs:oldisr] ; call the original ISR
    ; iret

start:
    xor ax, ax
    mov es, ax ; point es to IVT base
    mov ax, [es:9*4]
    mov [oldisr], ax ; save offset of old routine
    mov ax, [es:9*4+2]
    mov [oldisr+2], ax ; save segment of old routine
    cli ; disable interrupts
    mov word [es:9*4], kbisr ; store offset at n*4
    mov [es:9*4+2], cs ; store segment at n*4+2
    sti ; enable interrupts
```

l1:

```
mov ah, 0 ; service 0 – get keystroke
int 0x16 ; call BIOS keyboard service
cmp al, 27 ; is the Esc key pressed
jne l1 ; if no, check for next key
mov ax, [oldisr] ; read old offset in ax
mov bx, [oldisr+2] ; read old segment in bx
cli ; disable interrupts
mov [es:9*4], ax ; restore old offset from ax
mov [es:9*4+2], bx ; restore old segment from bx
sti ; enable interrupts
mov ax, 0x4c00 ; terminate program
int 0x21
```



Best of luck